MACHINE WHITE PAPERS

Hwacheon Hi-Eco 21 HS — Bearkat Blur

High-speed CNC lathe for small-part precision turning where accuracy and cycle time matter most. 8" OD × 12" L max turning.

Bearkat Blur · high-speed precision CNC turning center

Bearkat Blur exists because certain oilfield and aerospace parts are small, high-quantity, and can't tolerate the setup time of a larger machine. Bushings, couplings, precision shafts, valve inserts, and any repeatable geometry in the 8-inch envelope — this machine cycles fast without giving up accuracy. When cycle time drives the quote, this is the machine that wins.

What Bearkat Blur is for

Bearkat Blur is the shop's answer to a specific economic problem: some oilfield and aerospace parts are small, wanted in quantity, and cannot carry the setup time of a larger lathe. 8" OD × 12" L, a 2" through-spindle bar capacity, 4,000 RPM, and a 12-station turret with 1,181 IPM rapids on both axes — every one of those numbers is about cycle time. When the quote is won or lost on price per piece at quantity 200, this is the machine that wins it.

Pick this machine when

- The part is under 8" in diameter and 12" long — bushings, couplings, valve inserts, precision shafts, seat rings.
- Quantity is high enough that seconds per piece show up in the price.
- Bar-fed work up to 2" through the spindle.
- The part is small but the tolerance isn't loose — this machine holds ±0.0005" while cycling fast.

> **When it's the wrong machine**
>
> Anything that doesn't fit. The envelope is a hard wall, and the honest answer on a 9-inch part is that it goes to a Hi-Eco 35 cell instead. Heavy interrupted roughing on forged stock is also wrong here — 15 HP continuous is sized for finish-critical small work, not for taking scale off a forging. That's the Hammer of Forge's job.

From the floor

Small parts in Inconel 718 and 17-4 PH punish tool wear more than large ones, because the same insert edge is doing proportionally more of the part. On repeat runs the insert-change interval gets set from measured wear on the first batch rather than from a catalogue number, and finish-critical features get a fresh edge regardless of where that interval lands.

Pick your industry

Hwacheon Hi-Eco 21 HS for Oil & Gas

Full spec sheet, typical oil & gas parts, alloys, tolerance expectations, documentation approach.

Hwacheon Hi-Eco 21 HS for Aerospace

Full spec sheet, typical aerospace parts, alloys, tolerance expectations, documentation approach.

Hwacheon Hi-Eco 21 HS for Military & Defense

Full spec sheet, typical defense parts, alloys,

Hwacheon Hi-Eco 21 HS for Industrial & General Manufacturing

Full spec sheet, typical industrial parts, alloys, tolerance expectations, documentation approach.

tolerance expectations, documentation approach.

Have a project?